

Lab 18 - expected output

Reference transcript for Lab 18 | PersistentVolumes, Claims and StorageClasses, namespace kcna-lab18.

How to read this file. UUIDs, timestamps, ages, Pod names, Pod IPs and node names are environment-specific and will differ. Object names, phases, capacities, access modes, reclaim policies, row counts and error strings must match exactly.

>

The VOLUMEATTRIBUTESCLASS column appears on Kubernetes v1.29+ clients and shows <unset>; on older clients it is absent. Either is fine.

Prerequisite survey - what kind gives you

```
$ kubectl get storageclass
NAME                                PROVISIONER            RECLAIMPOLICY    VOLUMEBINDINGMODE
ALLOWVOLUMEEXPANSION    AGE
standard (default)    rancher.io/local-path    Delete            WaitForFirstConsumer    false
                                41m

$ kubectl -n local-path-storage get pods
NAME                                READY    STATUS    RESTARTS    AGE
local-path-provisioner-7dc846544d-mrxsq    1/1      Running    0            41m
```

Required signal: exactly one class, named standard, marked (default), provisioner rancher.io/local-path, Delete, WaitForFirstConsumer. If your cluster shows a different default class the rest of the lab still works, but Step 9's Pending behaviour depends on WaitForFirstConsumer - check your class's binding mode before assuming a fault.

```
$ kubectl get csidrivers
No resources found
```

Empty is expected on stock kind: local-path-provisioner predates CSI and is not a CSI driver.

Step 1 - datasets

```
$ echo "rows: $(( $(wc -l < data/manifest-archive.csv) - 1 ))"
rows: 30

$ kubectl -n kcna-lab18 create configmap tracklane-archive-seed --from-file=manifest-archive.csv=data/manifest-archive.csv
configmap/tracklane-archive-seed created
$ kubectl -n kcna-lab18 create configmap tracklane-retention-policy --from-file=retention-policy.csv=data/retention-policy.csv
configmap/tracklane-retention-policy created
```

Step 3 - static PV

```
$ kubectl get pv tracklane-archive-pv
```

NAME	CAPACITY	ACCESS MODES	RECLAIM POLICY	STATUS	CLAIM
tracklane-archive-pv	1Gi	RWO	Retain	Available	kcna-lab18/archive-claim
archive-claim	tracklane-manual	<unset>		6s	

Required signal: STATUS=Available and CLAIM already populated from claimRef before any PVC exists.

```
$ kubectl get pv tracklane-archive-pv -o jsonpath='{.metadata.namespace}{ "[end]"} {"\n"}' [end]
```

Empty before [end] proves the object is cluster-scoped.

Step 4 - immediate static binding

```
$ kubectl -n kcna-lab18 get pvc archive-claim
```

NAME	STATUS	VOLUME	CAPACITY	ACCESS MODES	STORAGECLASS
archive-claim	Bound	tracklane-archive-pv	1Gi	RWO	tracklane-manual
		<unset>			4s

Required signal: Bound within a few seconds (no WaitForFirstConsumer here - tracklane-manual names no StorageClass, so there is no binding mode to honour), and CAPACITY=1Gi despite a 512Mi request.

```
$ kubectl get pv tracklane-archive-pv -o custom-columns='NAME:.metadata.name,STATUS:.status.phase,CLAIM:.spec.claimRef.name,UID:.spec.claimRef.uid'
```

NAME	STATUS	CLAIM	UID
tracklane-archive-pv	Bound	archive-claim	1c9f4a02-6e5b-4d81-93a7-0b2e5f7c4d16

The UID is populated by the controller and will differ on your cluster.

Step 5 - seeding the volume

```
$ kubectl -n kcna-lab18 logs archive-seeder
wrote 30 archive rows to the PersistentVolume
sealed_by=archive-seeder
sealed_utc=2026-09-05T05:02:41Z
archive_rows=30
retention_class=gold
```

30 must match wc -l data/manifest-archive.csv minus one. sealed_utc varies.

```
$ kubectl -n kcna-lab18 delete pod archive-seeder --wait=true
pod "archive-seeder" deleted
$ kubectl -n kcna-lab18 get pods
No resources found in kcna-lab18 namespace.
```

Step 6 - persistence proof

```
$ kubectl -n kcna-lab18 logs archive-auditor
--- provenance written by the (now deleted) seeder Pod ---
sealed_by=archive-seeder
sealed_utc=2026-09-05T05:02:41Z
archive_rows=30
retention_class=gold
--- archive row count as read back ---
rows_read=30
--- can this auditor modify the archive? ---
read-only mount refused the write (expected)
```

This is the core result of the lab. sealed_by=archive-seeder and sealed_utc are byte-identical to Step 5, read by a Pod that did not write them. rows_read=30.

Step 8 - second StorageClass

```
$ kubectl get storageclass
NAME                                PROVISIONER                RECLAIMPOLICY  VOLUMEBINDINGMODE
ALLOWVOLUMEEXPANSION  AGE
standard (default)    rancher.io/local-path    Delete          WaitForFirstConsumer  false
                    52m
tracklane-retain     rancher.io/local-path    Retain          WaitForFirstConsumer  false
                    5s
```

Same provisioner, different RECLAIMPOLICY. That contrast is the point.

Step 9 - the expected Pending

```
$ kubectl -n kcna-lab18 get pvc render-cache
NAME          STATUS  VOLUME  CAPACITY  ACCESS MODES  STORAGECLASS
VOLUMEATTRIBUTESCLASS  AGE
render-cache  Pending                                standard      <unset>
8s

$ kubectl -n kcna-lab18 describe pvc render-cache | tail -8
Used By:          <none>
Events:
  Type    Reason              Age           From                    Message
  ----    -
  Normal  WaitForFirstConsumer  3s (x2 over 10s)  persistentvolume-controller  waiting for
first consumer to be created before binding
```

Required signal: Type: Normal, Reason: WaitForFirstConsumer. This is not a defect. Repeat counts grow while you wait.

Step 10 - binding on first consumer

```
$ kubectl -n kcna-lab18 rollout status deployment/render-cache --timeout=180s
deployment "render-cache" successfully rolled out

$ kubectl -n kcna-lab18 get pvc render-cache
```

NAME	STATUS	VOLUME	CAPACITY	ACCESS MODES
STORAGECLASS	VOLUMEATTRIBUTESCLASS	AGE		
render-cache	Bound	pvc-8b2e17d4-05c9-4f36-a1b7-e39d602c8a54	256Mi	RWO
standard	<unset>	2m41s		

Required signal: Bound, and the volume name begins pvc- followed by a UUID (yours will differ).

```
$ kubectl -n kcna-lab18 logs deployment/render-cache
--- /cache/starts.log ---
start render-cache-6c9f84b57d-x2klp 2026-09-05T05:14:08Z
--- bronze-tier datasets from the retention policy ---
tracking-render-cache
lane-etl-scratch
```

The two bronze-tier dataset names come from data/retention-policy.csv and must appear exactly.

After rollout restart:

```
$ kubectl -n kcna-lab18 logs deployment/render-cache | head -4
--- /cache/starts.log ---
start render-cache-6c9f84b57d-x2klp 2026-09-05T05:14:08Z
start render-cache-7fb5d9c684-qn8vt 2026-09-05T05:15:52Z
```

Required signal: two start lines with different Pod names. One line means the volume was re-provisioned rather than reused - check the PVC did not get recreated.

Step 12 - Retain, Released and manual re-admission

```
$ kubectl get pv tracklane-archive-pv
```

NAME	STATUS	VOLUME	CAPACITY	ACCESS MODES	RECLAIM POLICY	CLAIM
tracklane-archive-pv	Released	1Gi	RWO	Retain		kcna-lab18/
archive-claim	tracklane-manual	<unset>			17m	

Required signal: Released, not Available and not gone.

Recreated claim will not bind:

```
$ kubectl -n kcna-lab18 get pvc archive-claim
```

NAME	STATUS	VOLUME	CAPACITY	ACCESS MODES	STORAGECLASS
VOLUMEATTRIBUTESCLASS	AGE				
archive-claim	Pending				tracklane-manual
	10s				<unset>

After clearing the stale claimRef.uid:

```
$ kubectl patch pv tracklane-archive-pv --type=json -p='[{"op":"remove","path":"/spec/claimRef/uid"}, {"op":"remove","path":"/spec/claimRef/resourceVersion"}]'
```

persistentvolume/tracklane-archive-pv patched

```
$ kubectl -n kcna-lab18 get pvc archive-claim
```

NAME	STATUS	VOLUME	CAPACITY	ACCESS MODES	STORAGECLASS
archive-claim	Bound	tracklane-archive-pv	1Gi	RWO	tracklane-manual
		VOLUMEATTRIBUTESCLASS AGE			
		<unset> 33s			

And the data is untouched:

```
$ kubectl -n kcna-lab18 logs archive-auditor
--- provenance written by the (now deleted) seeder Pod ---
sealed_by=archive-seeder
sealed_utc=2026-09-05T05:02:41Z
...
rows_read=30
```

sealed_utc must equal the value from Step 5 - that is the proof the release/re-admit cycle preserved the bytes.

Graded checks

```
$ bash verification/checks.sh
== Lab 18 verification - namespace kcna-lab18 ==
[PASS] namespace kcna-lab18 exists
[PASS] PersistentVolume tracklane-archive-pv exists and is cluster-scoped
[PASS] tracklane-archive-pv capacity=1Gi accessMode=ReadWriteOnce reclaim=Retain
[PASS] tracklane-archive-pv is pre-bound via claimRef to kcna-lab18/archive-claim
[PASS] pvc/archive-claim is Bound to tracklane-archive-pv
[PASS] archive-claim requested 512Mi but was granted the PV's full 1Gi
[PASS] the archive on the PV holds 30 rows (matches data/manifest-archive.csv)
[PASS] PROVENANCE on the PV was written by a Pod that no longer exists
[PASS] storageclass tracklane-retain exists with reclaimPolicy=Retain
[PASS] tracklane-retain uses volumeBindingMode=WaitForFirstConsumer
[PASS] pvc/render-cache is Bound to a dynamically provisioned pvc-* volume
[PASS] the dynamic PV inherited reclaimPolicy=Delete from storageclass standard
[PASS] deployment/render-cache is available and mounts pvc/render-cache
[PASS] /cache/starts.log survived a rollout restart (2 start records)
-----
14 passed, 0 failed
```

Exit code 0.

Failure injection - StorageClass name typo

```
$ kubectl -n kcna-lab18 describe pvc archive-claim-typo
Name:          archive-claim-typo
Namespace:     kcna-lab18
StorageClass:  tracklane-manul
Status:        Pending
Volume:
Finalizers:    [kubernetes.io/pvc-protection]
Used By:       <none>
Events:
  Type            Reason              Age             From                                     Message
  ----            -
  Warning         ProvisioningFailed   4s (x3 over 18s) persistentvolume-controller             storageclass.storage.k8s.io "tracklane-manul" not found
```

Required signal: Type: Warning, Reason: ProvisioningFailed, and the literal string storageclass.storage.k8s.io "tracklane-manul" not found. Contrast with the Normal / WaitForFirstConsumer in Step 9 - that distinction is the graded diagnostic skill.

Immutability proof:

```
$ kubectl -n kcna-lab18 patch pvc archive-claim -p '{"spec":{"storageClassName":"standard"}}'
The PersistentVolumeClaim "archive-claim" is invalid: spec: Forbidden: spec is immutable after creation except resources.requests and volumeAttributesClassName for bound claims
```

The exact wording of the "except ..." clause varies slightly by minor version; the Forbidden: spec is immutable prefix is stable.

Cleanup

```
$ kubectl delete namespace kcna-lab18 --wait=true
namespace "kcna-lab18" deleted

$ kubectl get pv | grep kcna-lab18
tracklane-archive-pv 1Gi RWX Retain Released kcna-lab18/archive-claim tracklane-manual <unset> 38m

$ kubectl delete persistentvolume tracklane-archive-pv
persistentvolume "tracklane-archive-pv" deleted

$ kubectl delete storageclass tracklane-retain
storageclass.storage.k8s.io "tracklane-retain" deleted

$ kubectl get storageclass
NAME                PROVISIONER          RECLAIMPOLICY  VOLUMEBINDINGMODE
ALLOWVOLUMEEXPANSION AGE
standard (default)  rancher.io/local-path Delete          WaitForFirstConsumer false
81m

$ kubectl get pv
No resources found
```

Required end state: only the original standard StorageClass remains, no PersistentVolumes remain, and kcna-lab18 is gone. The dynamic PV for render-cache disappeared on its own because its class reclaim policy is Delete; tracklane-archive-pv survived because its policy is Retain and had to be removed by explicit name.